SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA0328	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	2023-2027	**Date:**	

Code of Conduct

I Keerthika S / 192371013 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Keerthika S

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Given: Spectrum = 45 MHz; one simplex channel = 25 kHz.
Simplex channels = 45 MHz ÷ 25 kHz = 45,000 kHz ÷ 25 kHz = 1800.
A full-duplex channel needs two simplex channels (one forward, one reverse).
Full-duplex channels = 1800 ÷ 2 = 900.
Ans: 900 full-duplex voice channels.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Given: Total channels = 560; cluster size N = 7.
In frequency reuse, channels are divided equally among the N cells of a cluster.
Channels per cell = 560 ÷ 7 = 80.
Ans: 80 channels per cell.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Given: 100 carriers (200 kHz spacing), 8 time slots per carrier.
Total voice channels = carriers × slots per carrier.
= 100 × 8 = 800.
Ans: 800 simultaneous voice channels.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

Given: Base station power = 80 W; handset power = 4 W.
Power ratio = 80 ÷ 4 = 20.
Ratio (dB) = 10·log10(20) = 10 × 1.301 = 13.01 dB.
Ans: ≈ 13 dB.

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Given: Carrier C = −60 dBm; Interference I = −92 dBm.
C/I (dB) = C(dBm) − I(dBm).
C/I = −60 − (−92) = 32 dB.
Ans: C/I = 32 dB (good link quality).

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Given: Spectrum = 42 MHz; channel width = 300 kHz; N = 7.
Total channels = 42 MHz ÷ 300 kHz = 42,000 ÷ 300 = 140.
Channels per cell = 140 ÷ 7 = 20.
Ans: 20 channels per cell.

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Given: Pr = 64 W at distance d; new distance = 2d; path-loss exponent n = 4.
Received power varies as Pr ∝ 1 / d^n.
New Pr = 64 × (1/2)^4 = 64 ÷ 16 = 4 W.
Ans: 4 W.

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Given: Radius doubles (R → 2R); inverse-square law (n = 2).
Received power ∝ Pt / R². To keep the same SNR at the new edge: Pt_new / (2R)² = Pt_old / R².
Pt_new = 4 × Pt_old.
Ans: Transmit power must be increased by a factor of 4 (≈ 6 dB).

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Given: Cluster size N = 4; 15 channels per cell.
Total channels in cluster = N × channels per cell.
= 4 × 15 = 60.
Ans: 60 channels.

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Given: v = 100 km/h = 27.78 m/s; f = 900 MHz; c = 3 × 10^8 m/s.
Doppler shift fd = v·f / c.
fd = (27.78 × 900 × 10^6) ÷ (3 × 10^8) ≈ 83.3 Hz.
Ans: ≈ 83 Hz (maximum Doppler shift).

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Given: Frame duration = 5 ms; slots per frame = 8.
Slot duration = frame duration ÷ number of slots.
= 5 ms ÷ 8 = 0.625 ms.
Ans: 0.625 ms (625 µs) per slot.

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Given: Bandwidth = 24 MHz; carrier spacing = 200 kHz.
Number of carriers = 24 MHz ÷ 200 kHz = 24,000 ÷ 200.
= 120.
Ans: 120 carrier frequencies.

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Given: Pt = 8 W; path loss = 33 dB.
Pt in dBm = 10·log10(8000 mW) = 39.03 dBm.
Pr = Pt − Path loss = 39.03 − 33 = 6.03 dBm.
Ans: ≈ 6 dBm (about 4 mW).

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Given: f = 1900 MHz; d0 = 1 m; d = 2.5 km = 2500 m; λ = c/f = 0.158 m.
Free-space loss at d0: PL(d0) = 20·log10(4π·d0/λ) = 20·log10(79.6) ≈ 38 dB.
PL(d) = PL(d0) + 20·log10(d/d0) = 38 + 20·log10(2500) = 38 + 68.
Ans: Free-space path loss ≈ 106 dB.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Given: 6 cells per cluster; 250 calls per cell.
Cluster capacity = 6 × 250.
Ans: 1500 simultaneous calls.

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Given: Call arrival rate λ = 4000 calls/s; average holding time H = 3 min = 180 s.
Active calls = λ × H (Little's law).

= 4000 × 180 = 720,000.
Ans: 720,000 calls active at one time.

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Given: 150 channels; 25 kHz per channel.
Total bandwidth = 150 × 25 kHz = 3750 kHz.
Ans: 3.75 MHz.

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Given: Guard band = 40 kHz; number of bands = 30.
Total wasted bandwidth = 30 × 40 kHz = 1200 kHz.
Ans: 1.2 MHz wasted in guard bands.

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Given: v = 90 km/h; cell radius = 1.5 km, so diameter = 3 km.
Distance travelled in 1 hour = 90 km.
Handoffs = distance ÷ cell diameter = 90 ÷ 3 = 30.
Ans: About 30 handoffs per hour.

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Given: 8 regional centers; 40 base stations each; 250 users per base station.
Total users = 8 × 40 × 250 = 320 × 250.
Ans: 80,000 users.

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Given: Area = 400 km^2; R = 400 m = 0.4 km; cell area ≈ R^2.
Cell area = $(0.4)^2$ = 0.16 km^2.
Number of cells = 400 ÷ 0.16 = 2500.
Ans: About 2500 cells.

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Given: Total spectrum = 6 MHz; channel width = 30 kHz.
Channels = 6 MHz ÷ 30 kHz = 6000 ÷ 30.
Ans: 200 channels.

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Given: 1200 subscribers; 4 calls/hour each; 2.5 min per call.
Calls per hour λ = 1200 × 4 = 4800; H = 2.5/60 hour.
Traffic A = λ × H = 4800 × (2.5 ÷ 60) = 200.
Ans: 200 Erlangs.

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Given: Bandwidth = 40 MHz; each user = 400 kHz.
Maximum users = 40 MHz ÷ 400 kHz = 40,000 ÷ 400.
(The 2.5 GHz carrier only fixes the band's position, not the user count.)
Ans: 100 users.

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Given: 600 users per base station; 1200 base stations.
Total capacity = 600 × 1200.
Ans: 720,000 users.

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Given: 4 cells paged at a time; 200 cells in the network.
Paging rounds (worst case) = 200 ÷ 4.
Ans: 50 paging rounds.

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Given: Capacity = 100,000 connections; 40,000 attempts/min; holding time = 3 min.
Offered traffic = 40,000 × 3 = 120,000 Erlangs.
Excess (blocked) = 120,000 − 100,000 = 20,000.
Blocked fraction = 20,000 ÷ 120,000 = 0.167.
Ans: ≈ 16.7% of offered traffic is blocked.

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Given: Subscribers = 600,000; roaming = 15%.
Roaming subscribers = 0.15 × 600,000.
Ans: 90,000 subscribers.

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Given: Capacity = 12,000 calls/s; current load = 9,000 calls/s.
Utilization = (9000 ÷ 12,000) × 100.
Ans: 75%.

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Given: Spectrum = 48 MHz; channel = 200 kHz; reuse factor N = 4.
Total channels = 48 MHz ÷ 200 kHz = 240.
Channels per cell = 240 ÷ 4 = 60.
Ans: 60 channels per cell.

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Given: Allocation = 30 MHz; channel spacing = 200 kHz.
Channels = 30 MHz ÷ 200 kHz = 30,000 ÷ 200.
Ans: 150 channels.

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Given: 2500 handoff attempts; success rate = 99%.
Successful handoffs = 0.99 × 2500.
Ans: 2475 calls.

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Given: Pt = 12 W; path loss = 45 dB.
Pt in dBm = 10·log10(12,000 mW) = 40.79 dBm.
Pr = 40.79 − 45 = −4.21 dBm.
Ans: ≈ −4.2 dBm.

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Given: Cluster = 7 cells; 900 users per cell.
Cluster capacity = 7 × 900.
Ans: 6300 users.

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Given: Frame = 4.8 ms; slots = 8.
Slot duration = 4.8 ms ÷ 8.
Ans: 0.6 ms (600 µs) per slot.

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Given: v = 150 km/h = 41.67 m/s; f = 1.8 GHz; λ = c/f = 0.1667 m.
Doppler shift fd = v/λ = v·f / c.
fd = 41.67 ÷ 0.1667 ≈ 250 Hz.
Ans: ≈ 250 Hz.

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Given: Capacity = 250,000 calls; arrival rate = 70,000/min; holding time = 4 min.
Offered load = 70,000 × 4 = 280,000 Erlangs.
Excess (blocked) = 280,000 − 250,000 = 30,000.
Blocked share = 30,000 ÷ 280,000 = 0.107.
Ans: ≈ 10.7% of the offered load is blocked.

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Given: Cluster size = 9 cells; 600 users per cell.
Cluster capacity = 9 × 600.
Ans: 5400 users.

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Given: 60,000 calls/s; drop rate = 0.4% = 0.004.
Dropped calls = 60,000 × 0.004.
Ans: 240 calls dropped per second.

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Given: Pt = 43 dBm; Pr = −87 dBm at the cell edge (2 km).
Path loss = Pt − Pr = 43 − (−87).
Ans: 130 dB.

Q41. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Given: Area per base station = 2.5 km²; stations = 150.
Total area = 2.5 × 150.
Ans: 375 km².

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Given: Spectrum = 48 MHz; channel = 200 kHz; reuse factor = 6.
Total channels = 48 MHz ÷ 200 kHz = 240.
Channels per cell = 240 ÷ 6 = 40.
Ans: 40 channels per cell.

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Given: 6000 users per cycle; total users = 150,000.
Paging cycles = 150,000 ÷ 6000.
Ans: 25 paging cycles.

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Given: Call duration = 9 min; one handoff roughly every 3 min.
Handoffs = 9 ÷ 3.
Ans: About 3 handoffs.

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Given: Signal = −85 dBm; noise floor = −115 dBm.
SNR (dB) = S(dBm) − N(dBm) = −85 − (−115).
Ans: SNR = 30 dB.

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Given: 250 users; 150 kbps each.
Minimum bandwidth (data rate) = 250 × 150 kbps = 37,500 kbps.
Ans: 37.5 Mbps.

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Given: 1500 subscribers; 4 calls/hour each; 2.5 min per call.
Calls per hour $\lambda = 1500 \times 4 = 6000$; $H = 2.5/60$ hour.
Traffic $A = \lambda \times H = 6000 \times (2.5 \div 60) = 250$.
Ans: 250 Erlangs.

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Given: 250 control channels; 12 kHz each.
Total bandwidth = 250 × 12 kHz = 3000 kHz.
Ans: 3 MHz.

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Given: 1500 connections; 600 kbps each.
Total throughput = 1500 × 600 kbps = 900,000 kbps.
Ans: 900 Mbps.

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

Given: Coverage per tower = πR^2 with R = 4 km; region = 350 km².
Area per tower = $\pi \times 4^2 = 50.27$ km².
Towers = 350 ÷ 50.27 = 6.96.
Ans: ≈ 7 towers.